VoltAI: An Integrated AI-Driven Platform for Smart Energy Forecasting, Anomaly Detection, and Reinforcement-Learning-Based Optimization in Households and Microgrids

Dinesh Acharya · Chinmayee Bhat · Chaithra J. Suvarna · Amoolya Shetty · Shabari Shedthi B*
Department of Computer Science and Engineering, NMAM Institute of Technology (NMAMIT), Nitte Deemed-to-be-University, Nitte – 574110, Karnataka, India
**Corresponding author: shabarishetty@nitte.edu.in*

Abstract — The rapid proliferation of rooftop photovoltaic (PV) systems, battery energy storage, electric vehicles, and smart metering infrastructure has transformed residential power networks into highly dynamic, data-intensive microgrids. Accurate short- and medium-term forecasting of household energy consumption and solar generation, continuous anomaly detection, and adaptive optimization of battery storage and energy routing are now indispensable for maintaining grid stability, reducing operational costs, and ensuring safe system operation. This paper presents VoltAI, a unified AI-driven platform that tightly integrates four key functional layers: (i) a three-stage hybrid demand-forecasting pipeline combining CNN+LSTM for temporal feature extraction, XGBoost+Prophet for nonlinear pattern modeling, and a gradient-boosting meta-learner that fuses the two outputs into a final prediction; (ii) a solar generation prediction module coupling CNN-extracted sky and satellite imagery features from Google Earth Engine with meteorological gradient-boosting regression; (iii) a dual-method anomaly detection subsystem using Isolation Forest for abrupt load spikes and autoencoders for gradual PV degradation; and (iv) a Deep Q-Network (DQN) reinforcement-learning controller that optimizes real-time battery scheduling and energy routing, explicitly incorporating forecast uncertainty and anomaly confidence into its state representation. Implemented as a FastAPI backend with a React dashboard and PostgreSQL time-series database, VoltAI is rigorously evaluated on household smart-meter data, NOAA weather records, and Google Earth Engine satellite imagery. The hybrid forecasting ensemble reduces RMSE by approximately 10–15% over single-model baselines, while the anomaly-aware DQN controller measurably lowers grid dependence and operating costs under realistic residential scenarios. By coupling forecasting, monitoring, and control within one modular framework, VoltAI delivers a scalable blueprint for intelligent and sustainable smart-energy management.

Keywords — *smart grids, energy forecasting, solar generation prediction, hybrid ensemble, CNN-LSTM, XGBoost, gradient boosting, anomaly detection, autoencoder, Isolation Forest, deep Q-network, reinforcement learning, microgrid optimization, Google Earth Engine.*

I. INTRODUCTION

The global transition toward distributed and renewable energy has introduced pronounced stochasticity into power systems at the household and microgrid scales. Rooftop photovoltaic installations, battery energy storage systems (BESS), electric vehicles, and advanced metering infrastructure collectively produce highly nonlinear, weather-dependent, and time-varying supply–demand dynamics that classical statistical approaches are ill-equipped to handle. Accurate short-term load and solar forecasting, real-time anomaly detection, and adaptive energy management are therefore critical enablers of grid stability, cost minimization, and operational safety for modern prosumers — entities that both consume and generate energy.

Traditional energy management systems rely on fixed heuristics and rule-based schedulers that cannot adapt to sudden weather shifts, load variability, or equipment faults. Although machine learning and deep learning techniques have individually demonstrated strong performance in each sub-domain — forecasting, anomaly detection, and optimization — the majority of deployed systems treat these as independent pipelines, creating fragmented workflows that prevent real-time feedback between components. In particular, reinforcement-learning (RL) controllers, which have shown promise for adaptive battery scheduling and energy routing, often degrade in performance when operating on stale or inaccurate forecasts, or when anomalous system states are not detected and communicated to the controller.

VoltAI addresses these limitations through a tightly integrated, modular AI platform. The most significant architectural advancement over prior work — including the authors' own earlier conference publication — is the redesigned demand-forecasting pipeline. Rather than a simple weighted average of individual model outputs, the updated system employs a three-stage hybrid architecture: (1) a CNN+LSTM stack extracts temporal and sequence-level features; (2) an XGBoost+Prophet ensemble captures nonlinear relationships and seasonal

decomposition; and (3) a gradient-boosting meta-learner is trained on the outputs of both stages to produce a final fused prediction. This stacked generalization approach consistently outperforms both individual models and naïve ensembles. Similarly, the solar generation prediction module has been updated to incorporate Google Earth Engine satellite imagery processed through a CNN, fused with meteorological gradient-boosting regression for improved sky-condition modeling.

The principal contributions of this paper are as follows:

- A three-stage hybrid demand forecasting architecture — CNN+LSTM → XGBoost+Prophet → Gradient Boosting meta-learner — delivering 10–15% RMSE reduction over single-model baselines.
- An updated solar generation prediction module combining CNN-extracted Google Earth Engine satellite imagery with meteorological gradient-boosting regression for improved intra-hour and day-ahead PV output estimates.
- A dual-method anomaly detection subsystem fusing Isolation Forest (for abrupt load spikes) and autoencoders (for gradual PV degradation), with anomaly confidence scores fed directly into the RL controller's state.
- A Deep Q-Network optimization agent whose state representation explicitly incorporates forecast uncertainty and anomaly scores, triggering a conservative safety fallback when either signal exceeds a reliability threshold.
- A scenario simulation interface enabling interactive what-if analysis across environmental and usage parameters.
- A production-grade full-stack deployment: FastAPI backend, PostgreSQL time-series database, and a React dashboard with integrated visualization and explainability components.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details the updated system architecture. Section IV describes the complete methodology including mathematical formulations. Section V presents experimental results and comparative analysis. Section VI discusses limitations. Section VII concludes.

II. RELATED WORK

A. Energy Demand Forecasting

Early short-term load forecasting research relied on classical statistical models such as ARIMA and SARIMAX, which assume linearity and stationarity — properties poorly satisfied by real-world energy time series exhibiting weather sensitivity, behavioral variability, and calendar effects. The transition to machine learning and deep learning has brought substantial improvements. Semmelmann et al. [1] demonstrated that a hybrid LSTM+XGBoost ensemble outperforms unimodal approaches for community-level smart-meter forecasting by exploiting complementary model strengths. Arslan [18] confirmed this finding across household datasets, showing that combining sequence-based models with gradient-boosting algorithms improves both accuracy and generalization. Qureshi et al. [27] and Maarif et al. [19] further showed that explainable LSTM architectures achieve competitive accuracy while improving user trust in prediction outputs. Albahli [7] provided a systematic comparison of LSTM and Prophet, finding that Prophet's interpretable trend-seasonality decomposition provides a useful baseline but is outperformed by deep learning under highly variable demand conditions.

B. Solar Generation Forecasting

The intermittent nature of solar irradiance due to cloud dynamics and atmospheric variability makes PV output prediction a particularly challenging regression task. Vision-based approaches using ground-based sky cameras and satellite imagery have gained significant traction. Paletta et al. [2] reviewed the state of the art in computer vision for solar forecasting, noting that CNN architectures applied to sky imagery excel at extracting cloud-motion and atmospheric transmission features relevant to intra-hour irradiance prediction. Lim et al. [3] implemented a CNN–LSTM architecture combining sky image embeddings with temporal modeling, achieving strong short-term PV forecast accuracy. Abumohsen et al. [5] and Benítez et al. [4] explored hybrid CNN+gradient-boosting approaches for day-ahead forecasting, with the latter additionally quantifying prediction uncertainty. The use of Google Earth Engine (GEE) as a satellite imagery source for solar potential mapping and

panel monitoring has been validated by Supe et al. [6] and Haz et al. [15]. Song et al. [14] extended these results to probabilistic ultra-short-term forecasting frameworks. VoltAI builds on these findings by incorporating GEE satellite imagery within a CNN-based feature extractor fused with meteorological gradient-boosting regression.

C. Anomaly Detection in Energy Systems

Maintaining grid reliability and detecting equipment faults early require robust anomaly detection capabilities. Autoencoder-based methods have proven effective for identifying gradual performance degradation in PV systems: high reconstruction error indicates deviations from the learned normal operating signature. Gong et al. [13] introduced memory-augmented autoencoders (MemAE) to tighten the normal pattern boundary, while Raihan et al. [23] demonstrated Bi-LSTM autoencoders for sequential anomaly detection. Li et al. [11] applied sequential conditional variational autoencoders specifically to PV fault identification. Park et al. [21] validated convolutional autoencoders for this task. For abrupt load anomalies, Isolation Forest [28] offers computationally efficient and parameter-lean detection without requiring labeled fault data. Atemkeng et al. [12] proposed label-assisted autoencoders to reduce false alarms using limited supervision. Burgos et al. [22] and Banik et al. [26] provide comprehensive taxonomies of anomaly detection approaches for smart grids. VoltAI adopts a dual-method fusion — Isolation Forest for point anomalies and autoencoders for gradual drift — to achieve high recall with reduced false-positive rates.

D. Reinforcement Learning for Energy Management

Reinforcement learning has emerged as a compelling paradigm for energy management, enabling agents to learn near-optimal control policies through environmental interaction rather than requiring explicit system models. Nakabi and Toivanen [8] demonstrated DQN effectiveness for microgrid energy scheduling. Muriithi [9] and Chen et al. [10] applied improved DQN variants to battery charging and load allocation. Kwon and Zhu [25] incorporated battery degradation models into the RL reward structure. Shojaeighadikolaei et al. [24] explored multi-agent RL for distributed energy resource coordination. A recurring limitation in RL-based energy management is performance degradation when forecast quality is poor or when the operating environment presents states not encountered during training. VoltAI addresses this by injecting forecast uncertainty and anomaly confidence signals directly into the DQN state, and by triggering a safe rule-based fallback policy when these signals indicate unreliable conditions.

E. Identified Gaps

A synthesis of existing literature reveals four persistent gaps that motivate this work. First, most systems treat forecasting, anomaly detection, and optimization as independent pipelines, forgoing the performance benefits of tight inter-module coupling. Second, hybrid forecasting architectures rarely exploit stacked generalization — using a meta-learner trained on base model outputs — which consistently improves ensemble accuracy. Third, few systems integrate real satellite imagery (e.g., GEE) into the solar prediction pipeline alongside meteorological features within a unified deployment. Fourth, RL controllers in energy management seldom incorporate explicit forecast reliability and anomaly confidence signals into their state representations, limiting their adaptability under uncertain operating conditions. VoltAI is designed to address all four gaps.

III. SYSTEM ARCHITECTURE

VoltAI is organized as a five-layer modular architecture in which data flows sequentially through ingestion, preprocessing, intelligence, API, and presentation layers. Figure 1 illustrates the updated system architecture, reflecting the revised hybrid forecasting pipeline and the integration of GEE satellite data. The four functional intelligence modules — demand forecasting, solar generation prediction, anomaly detection, and RL-based optimization — share a common feature store and communicate through well-defined internal interfaces, enabling forecast outputs to propagate into the optimization module's state representation and anomaly scores to gate both the optimization policy and the user-facing alert system.

A. Data Ingestion and Storage Layer

Four primary data streams are ingested: (i) smart-meter records at 15-minute resolution (e.g., UCI Household Power Consumption dataset), (ii) meteorological variables from NOAA or local APIs (temperature, humidity, cloud cover, wind speed, solar irradiance), (iii) hourly sky and satellite imagery from Google Earth Engine, and

(iv) time-of-use grid price signals. All streams are aligned to a common 15-minute or hourly temporal grid. A PostgreSQL relational database stores preprocessed time-series records and all module outputs — forecast results, anomaly logs, optimization decisions, and simulation outputs — with timestamp-based indexing to support efficient time-range queries.

B. Preprocessing and Feature Engineering Layer

Missing values are imputed using forward-fill for gaps of up to three timesteps and linear interpolation for longer spans. Outliers are flagged using z-score thresholding ($|z| > 3.5$) and replaced with interpolated values prior to model training. Temporal features engineered include: hour of day, day of week, month, season, and binary holiday indicators. Rolling statistical aggregates — mean and standard deviation over 24-hour and 168-hour sliding windows — are computed to provide contextual history. All numeric features are standardized to zero mean and unit variance using statistics computed on the training split only, preventing data leakage. Sky and satellite images are resized to 128×128 pixels and normalized to [0, 1] per channel before CNN processing.

C. Intelligence Layer — Four Integrated Modules

The intelligence layer comprises four tightly integrated subsystems described in detail in Section IV. The critical architectural update relative to the prior conference paper is the demand-forecasting module, which now employs a three-stage stacked generalization approach rather than a simple weighted ensemble. The solar prediction module has been extended with GEE satellite imagery as an additional CNN input source. The anomaly detection and RL optimization modules retain their dual-method and DQN architectures respectively, with the addition of an explicit safety supervisor that gates the RL policy based on forecast and anomaly reliability signals.

D. API and Presentation Layers

A FastAPI REST API layer exposes all system capabilities as endpoint services with JWT-based authentication, HTTPS encryption, and support for both real-time inference and batch processing. A React-based web dashboard integrates Plotly visualization components for real-time energy flow graphs, 24-hour forecast overlays, anomaly alert markers, optimization recommendations, and the scenario simulation interface. XGBoost feature importance rankings and Prophet trend-seasonality decomposition are surfaced within the dashboard to provide explainable AI transparency for non-technical users.

IV. METHODOLOGY

A. Energy Demand Forecasting — Three-Stage Hybrid Architecture

The updated demand forecasting pipeline implements stacked generalization across three stages, as shown in Figure 1. This architecture supersedes the simple weighted ensemble used in the prior conference paper and consistently delivers superior accuracy by training a meta-learner on base model outputs rather than relying on pre-specified weights.

1) Stage 1 — CNN+LSTM Temporal Feature Extractor

The first stage employs a CNN+LSTM stack to capture both local temporal patterns and long-range sequence dependencies. One-dimensional convolutional layers with kernel size 3 and 64 filters slide over input sequences of length 168 (one week of hourly observations), producing feature maps that encode short-term consumption shape. These feature maps are passed to a two-layer LSTM whose gated memory cells selectively retain long-range dependencies across the input sequence. The LSTM hidden-state update equations are:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (1)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (2)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (3)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (4)$$

$$h_t = o_t \odot \tanh(c_t) \quad (5)$$

where f_t, i_t, o_t are forget, input, and output gates; c_t is the cell state; h_t is the hidden output; and W, U, b are learnable parameters. The final hidden state h_T is passed through a dense output layer to produce Stage 1 predictions $\hat{y}^{(CNN-LSTM)}$. Training minimizes the Huber loss for robustness to outliers:

$$L_{forecast} = (1/N) \Sigma_{i=1}^{N} Huber(y_i - \hat{y}_i) \quad (6)$$

2) Stage 2 — XGBoost + Prophet Nonlinear Pattern Modeler

The second stage runs in parallel with Stage 1. XGBoost is trained on the full tabular feature set — lag variables, rolling aggregates, temporal encodings, and meteorological covariates — to capture nonlinear feature interactions not easily modeled by recurrent architectures. Prophet operates independently on the raw time series, decomposing it into trend, weekly seasonality, annual seasonality, and holiday components to produce an interpretable baseline prediction. Both models yield Stage 2 predictions $\hat{y}^{(XGB)}$ and $\hat{y}^{(Prophet)}$ respectively.

3) Stage 3 — Gradient Boosting Meta-Learner

The meta-learner receives the Stage 1 and Stage 2 outputs — $[\hat{y}^{(CNN\text{-}LSTM)}, \hat{y}^{(XGB)}, \hat{y}^{(Prophet)}]$ — as input features and is trained on a held-out validation set to predict the true demand values, learning optimal non-linear combinations of the base predictions. A gradient-boosting regressor is used as the meta-learner for its robustness to correlation between base predictions and its built-in regularization. The final ensemble prediction is:

$$\hat{y}_h = f_{GB}([\hat{y}_h^{CNN\text{-}LSTM}, \hat{y}_h^{XGB}, \hat{y}_h^{Prophet}]; \theta_{meta}) \quad (7)$$

This stacked architecture substantially outperforms a fixed weighted average (used in the prior conference implementation) because the meta-learner adapts its combination weights based on input context rather than applying fixed global weights.

B. Solar Generation Prediction

The solar prediction module combines two information sources — sky/satellite imagery and meteorological data — which are independently processed and then fused, as illustrated in the architecture diagram.

1) CNN Image Feature Extractor

Hourly satellite images obtained from Google Earth Engine are processed by a CNN. The convolution operation applied to input image I with kernel K is:

$$(I * K)(x, y) = \Sigma_u \Sigma_v I(x+u, y+v) \cdot K(u, v) \quad (8)$$

Multiple convolutional and max-pooling layers extract progressively abstract representations of sky conditions — cloud coverage, cloud type, irradiance transmission characteristics — producing a compact image embedding z_{img}. Batch normalization is applied after each convolutional block to stabilize training.

2) Meteorological Gradient Boosting

Meteorological variables (temperature, humidity, cloud cover fraction, wind speed, clearness index) are processed by a gradient-boosting regressor, yielding a meteorological embedding z_{met} that captures relationships between weather features and PV output that may not be visually apparent in imagery alone.

3) Multimodal Fusion

Both embeddings are concatenated and passed through a multi-layer perceptron (MLP) fusion network:

$$\hat{p}_t = f_{merge}(z_{img}, z_{met}; \theta) \quad (9)$$

where $\hat{p}_t$ is the predicted solar power at time t. This architecture allows the system to leverage complementary information — visual cloud features from imagery and numerical meteorological variables — in a unified prediction, improving accuracy particularly under rapidly changing sky conditions.

C. Anomaly Detection — Dual-Method Fusion

The anomaly detection subsystem uses two complementary methods targeting different anomaly types that manifest differently in load and PV data.

1) Isolation Forest for Abrupt Load Anomalies

Isolation Forest [28] identifies abrupt load spikes and sudden consumption deviations by measuring how easily individual samples can be isolated from the rest of the feature space through random partitioning. Samples requiring fewer partitions — shorter average path lengths — are assigned higher anomaly scores. This unsupervised approach requires no labeled fault data and is computationally efficient at inference time, making it well-suited for real-time load monitoring.

2) Autoencoder for Gradual PV Degradation

A deep autoencoder is trained on normal PV operating sequences to learn a compact latent representation of healthy system behavior. At inference time, the reconstruction error:

$$L_AE = \|x - \tilde{x}\|_2^2 \quad (10)$$

serves as the anomaly score. Progressive increases in reconstruction error over multiple consecutive timesteps indicate gradual PV degradation phenomena — panel soiling, inverter drift, shading — that manifest as slow deviations from the learned normal signature. MemAE and Bi-LSTM autoencoder variants are additionally evaluated for complex temporal anomaly patterns.

3) Score Fusion and Confidence Signal

Isolation Forest and autoencoder anomaly scores are independently normalized to [0, 1] and combined as:

$$score_fused = \alpha \cdot score_AE + (1 - \alpha) \cdot score_IF, \quad \alpha = 0.6 \quad (11)$$

Samples with score_fused exceeding a threshold τ (set at the 99th percentile of training-set scores) are flagged as anomalies. The fused score is also exported as a continuous anomaly confidence signal, which is incorporated into the DQN state representation described below.

D. Deep Q-Network Energy Optimization

Energy routing between the grid, battery, and solar source is optimized by a reinforcement learning agent built on the Deep Q-Network (DQN) framework. The energy management problem is formulated as a discrete-time Markov Decision Process.

1) State, Action, and Reward Formulation

The state vector at time t is:

$$s_t = [SoC_t, \hat{y}_\{t:t+24\}, \hat{p}_\{t:t+24\}, C_grid(t), ToD_t, \sigma_forecast(t), score_fused(t)] \quad (12)$$

where SoC_t is the battery state of charge, ŷ_{t:t+24} and p̂_{t:t+24} are the 24-hour demand and solar forecasts, C_grid(t) is the current grid price, ToD_t is a time-of-day encoding, σ_forecast(t) is the forecast uncertainty estimate, and score_fused(t) is the anomaly confidence signal. Explicitly including forecast uncertainty and anomaly confidence enables the controller to modulate its behavior based on input reliability — a key architectural improvement over the conference paper version.

Actions are discrete: {charge at rate r_c, discharge at rate r_d, idle} × {buy from grid, sell to grid, self-consume}. The reward function is:

$$r_t = -C_grid(t) \cdot E_grid(t) - \lambda_anomaly \cdot \mathbb{1}[score_fused > \tau] + \eta \cdot E_solar_used(t) \quad (13)$$

balancing grid cost minimization, anomaly safety penalties, and renewable energy utilization.

2) DQN Training

The DQN loss function minimizes the temporal-difference error:

$$L_DQN(\theta) = E_\{(s,a,r,s')\sim D\} [(r + \gamma \max_\{a'\} Q_\{\theta^-\}(s', a') - Q_\theta(s, a))^2] \quad (14)$$

where θ^- are the periodically-updated target network parameters and D is the experience replay buffer. Domain randomization over weather conditions and consumption patterns during simulation training improves policy robustness to distributional shift at deployment.

3) Safety Supervisor

A safety supervisor monitors σ_forecast(t) and score_fused(t) in real time. If either exceeds a pre-configured reliability threshold τ_safe, the learned RL policy is overridden by a conservative rule-based fallback that maintains the battery within safe state-of-charge bounds, avoids unnecessary grid transactions, and prioritizes direct solar self-consumption. This safety layer prevents the agent from making potentially harmful decisions when its input signals are unreliable.

E. Scenario Simulation Module

An interactive simulation interface allows users to modify environmental and usage parameters — temperature, appliance load level, solar panel tilt angle, and weather condition — via slider and dropdown controls. The system propagates these inputs through the forecasting and solar prediction modules to generate updated 24-hour energy profiles, estimated efficiency metrics, daily cost projections, and carbon savings estimates. This what-if analysis capability supports proactive energy planning and enables users to evaluate operational configurations before physical implementation.

F. Evaluation Metrics

Forecasting performance is quantified by MAE, RMSE, MAPE, and MSE. Anomaly detection performance is evaluated by Precision, Recall, F1-Score, and PR-AUC on a held-out labeled test set containing known fault and anomaly events. Optimization performance is assessed by total energy cost, grid-dependence ratio, hazardous events avoided, and anomaly prevention rate over a 30-day simulated test period.

V. RESULTS AND DISCUSSION

A. Energy Demand Forecasting Performance

Table I presents the forecasting performance of individual models and the updated three-stage hybrid ensemble on the held-out test set using household smart-meter data.

Method	RMSE	MAE	MAPE (%)	MSE
LSTM (standalone)	1.12	0.86	7.9	1.25
Prophet (standalone)	1.28	0.94	8.6	1.64
XGBoost (standalone)	1.05	0.81	7.1	1.10
CNN+LSTM (Stage 1)	1.01	0.79	6.9	1.02
XGBoost+Prophet (Stage 2)	0.99	0.77	6.7	0.98
Hybrid Ensemble — VoltAI (Stages 1+2+GB)	0.93	0.72	6.2	0.86

TABLE I: Energy demand forecasting performance. Lower is better. VoltAI hybrid (Stages 1+2+GB) achieves 10–17% RMSE improvement over single-model baselines.

The three-stage stacked generalization architecture consistently outperforms all single-model and partial ensemble configurations. The gradient-boosting meta-learner (Stage 3) provides additional accuracy improvement over the Stage 1 and Stage 2 outputs individually by learning optimal context-dependent combination strategies. The hybrid ensemble also demonstrates stronger generalization across seasonal transitions and periods of high demand variability, which are the primary failure modes of single-model approaches. Short-term (6-hour ahead) predictions achieve lower error than day-ahead (24-hour) forecasts, consistent with increasing uncertainty over longer horizons. Crucially, the updated three-stage pipeline outperforms the simple weighted ensemble from the earlier conference implementation on all metrics, validating the architectural redesign.

B. Solar Generation Prediction Performance

The updated solar prediction module — combining CNN-extracted GEE satellite imagery features with meteorological gradient-boosting regression — achieves improved intra-hour and day-ahead PV output estimates compared to weather-only or image-only baselines. CNN image features are most informative for intra-hour forecasting (cloud-motion features with short predictive horizon), while meteorological variables provide longer-range context. The fusion architecture via an MLP effectively combines both information sources, with the merged model achieving approximately 8–12% RMSE improvement over the best single-source baseline. These results are consistent with findings by Paletta et al. [2] and Lim et al. [3], and extend them to the GEE satellite imagery domain as established by Supe et al. [6].

C. Anomaly Detection Results

Table II summarizes the anomaly detection performance of individual methods and the fused approach on a labeled test set containing both abrupt load spikes and gradual PV degradation events.

Method	Precision	Recall	F1-Score	PR-AUC
Isolation Forest only	0.81	0.74	0.77	0.79
Autoencoder only	0.76	0.83	0.79	0.81
Fused (α=0.6) — VoltAI	0.88	0.86	0.87	0.89

TABLE II: Anomaly detection performance comparison. Fused approach outperforms individual methods on all metrics.

The dual-method fusion achieves the best F1-Score and PR-AUC, outperforming either method in isolation by 8–10 percentage points. Isolation Forest demonstrates high precision for point anomalies — sudden consumption spikes — while the autoencoder achieves higher recall for gradual degradation events, where its reconstruction error accumulates progressively before threshold crossing. The fusion strategy balances these complementary strengths, importantly reducing the false-alarm rate that would impair practical operational usability. Anomaly confidence scores are continuously forwarded to the DQN controller, enabling real-time adaptive response to detected faults.

D. DQN Optimization Results

The DQN optimization policy, trained over approximately 800 simulation episodes with domain randomization, converges to a stable control strategy that significantly outperforms a rule-based baseline (charge when solar surplus > 20%, discharge to meet demand when SoC > 30%). Over the 30-day simulated test period, the DQN achieves a measurable reduction in total energy cost relative to the baseline by better aligning battery discharge with peak grid pricing periods, reducing unnecessary grid imports during high solar generation, and avoiding battery over-discharge in advance of forecast high-demand periods. The explicit inclusion of forecast uncertainty and anomaly confidence in the state representation is confirmed to improve controller robustness: in test episodes with degraded forecast quality, the safety supervisor correctly triggers the fallback policy, preventing potentially costly or unsafe battery operations.

E. System Integration and Computational Performance

API response latency for single-step demand and solar forecasting requests is within acceptable real-time processing bounds for operational energy management (sub-second). Batch 24-hour forecasting completes efficiently enough for near real-time operational scheduling. The PostgreSQL backend handles datasets of hundreds of thousands of time-series records without significant query performance degradation. The React dashboard renders forecast overlays, anomaly alerts, optimization recommendations, and simulation outputs concurrently without perceptible latency under tested concurrent user loads.

F. Comparative Analysis with Traditional Methods

Compared to ARIMA-based baselines, VoltAI's hybrid ensemble achieves substantially lower prediction error, particularly under conditions of rapid demand variation and cloud-induced solar intermittency. Unlike standalone deep learning deployments, the integrated anomaly detection and optimization modules allow the system to respond to detected faults by adjusting energy management strategies dynamically — a capability absent from most research implementations. The full-stack deployment with secure API access and an interactive dashboard represents a step beyond experimental configurations reported in most related work. The conference paper version used individual model comparison without a meta-learner; the stacked generalization update delivers consistent improvements across all evaluation metrics.

G. Visualization and Explainability

The VoltAI dashboard presents actual-versus-predicted energy flow comparisons, 24-hour forecast overlays, anomaly alert timelines, AI optimization recommendations, and scenario simulation outputs in a unified interface. XGBoost feature importance rankings are displayed to communicate which input variables most influence current predictions. Prophet's trend-seasonality decomposition provides an intuitive visual explanation of forecast

structure. Informal user evaluation confirmed that these explainability components meaningfully improved user comprehension of model outputs, supporting more informed energy management decisions without requiring deep machine learning expertise.

VI. LIMITATIONS AND FUTURE WORK

While VoltAI demonstrates strong and consistent performance across all evaluated dimensions, several limitations warrant acknowledgment and motivate future work. The current implementation relies on historical datasets for model training and GEE satellite imagery accessed in batch mode; real-time IoT integration for live smart-meter streaming and real-time satellite data access remains a planned enhancement. The DQN optimization policy was trained and evaluated in a simulated environment using parameterized battery models; field trials with physical battery systems will require additional reward shaping to account for real battery degradation curves, temperature effects, and grid regulation requirements. The explainability components, while present, could be further enriched through formal SHAP (SHapley Additive exPlanations) integration for the LSTM and CNN models, complementing the existing XGBoost feature importance.

Priority future development directions include: (1) real-time IoT data streaming via Kafka/MQTT for live energy monitoring; (2) ONNX model export and FastAPI optimization for low-latency edge deployment; (3) extension to wind energy and hybrid renewable system modeling; (4) multi-agent RL for coordinated management of distributed energy resources across multiple households; (5) formal SHAP-based explainability for deep learning components; (6) integration with demand-response market APIs for cost-aware optimization and energy trading; (7) field validation of the complete system in real residential and community microgrid deployments.

VII. CONCLUSION

This paper presented VoltAI, a unified AI-driven smart energy management platform that integrates four tightly coupled functional modules: a three-stage hybrid demand forecasting pipeline (CNN+LSTM → XGBoost+Prophet → Gradient Boosting meta-learner), a multimodal solar generation prediction system combining GEE satellite imagery with meteorological gradient-boosting regression, a dual-method anomaly detection subsystem fusing Isolation Forest and autoencoders, and a DQN reinforcement-learning optimizer whose state explicitly incorporates forecast uncertainty and anomaly confidence.

The three-stage hybrid forecasting architecture — the key architectural advancement over the prior conference paper — achieves 10–17% RMSE improvement over single-model baselines and consistently outperforms the simple weighted ensemble of the earlier implementation. The dual-method anomaly detection fusion delivers 8–10 percentage point F1-Score improvements over individual methods by exploiting complementary sensitivities to abrupt and gradual anomaly types. The DQN optimization controller demonstrates measurably lower energy cost and grid dependence compared to rule-based baselines, with the safety supervisor providing reliable fallback behavior under degraded forecast or anomaly confidence conditions.

The full-stack deployment — FastAPI backend, PostgreSQL time-series database, React dashboard with integrated explainability — demonstrates that VoltAI is not only academically rigorous but practically deployable for smart grid, community energy management, and residential microgrid applications. By coupling forecasting, monitoring, and control within a single modular framework, VoltAI advances the state of the art in intelligent energy management and provides a scalable, interpretability-aware blueprint for sustainable smart-energy systems.

DISCLOSURES

The authors declare no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

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